UNIVERSITI TENAGA NASIONAL

The Energy University

Subject: Text Analytics (CISB5123)

Project title: Sentiment Analysis of Guest Reviews for Jabal Omar Marriott Hotel, Makkah Using Tripadvisor Data

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Table of Contents

1. **Real-World Text Analytics Problem 1**
2. **Project Objectives 2**
3. **Project Scope 3**
   3.1 Scope Includes
   3.2 Scope Excludes

4. **Dataset Description 4**

5. **Data Collection Method 5**

6. **Sampling 6**

7. **Text Preprocessing 7**
   7.1 Lowercasing
   7.2 URL Removal
   7.3 Removing Punctuation
   7.4 Removing Numbers
   7.5 Removing Extra Spaces
   7.6 Stop Word Removal
   7.7 Lemmatization
   7.8 Tokenization

8. **Ethical Considerations 8**

9. **Dataset Source 9**

1. Real-World Text Analytics Problem

Online travel platforms such as Tripadvisor allow hotel guests to share their experiences by writing detailed reviews about their stay. These reviews contain valuable information about customer satisfaction, hotel services, facilities, and overall guest experiences. However, when a hotel receives hundreds or thousands of reviews, it becomes difficult and time-consuming to manually read and analyze all the feedback.

Hotels may struggle to identify common complaints, positive experiences, and overall guest sentiment when customer opinions are scattered across many reviews. Without proper analysis, important insights into service quality, cleanliness, staff behavior, and location advantages may be overlooked.

To address this challenge, this project applies text analytics techniques to analyze guest reviews of Jabal Omar Marriott Hotel in Makkah collected from Tripadvisor. By processing and cleaning the textual review data, the project aims to extract meaningful insights about customer opinions in a structured and systematic way. This will help identify general sentiment trends and common topics mentioned by guests, which can assist hotel management in understanding customer feedback and improving their services.

2. Project Objectives

• To collect guest reviews from Tripadvisor for Jabal Omar Marriott Hotel, Makkah.
• To preprocess and clean the textual review data.
• To analyze overall customer sentiment.
• To identify common topics mentioned by guests such as service, cleanliness, and location.

3. Project Scope

This project focuses on analyzing textual guest reviews collected from Tripadvisor for one hotel: Jabal Omar Marriott Hotel, Makkah.

3.1Scope Includes

• Using English-language review texts as the main dataset
• Collecting reviews from Tripadvisor as the data source
• Performing text preprocessing such as:

- lowercasing
- removing punctuation
- tokenization
- stop-word removal

• Analyzing general sentiment trends and common topics mentioned by guests (e.g., service, cleanliness, location, facilities)

3.2Scope Excludes

- Comparing multiple hotels
- Analyzing images or ratings only
- Advanced deep learning models
- Non-English text analysis

4. Dataset Description

The dataset used in this project consists of guest reviews collected from **Tripadvisor** for the **Jabal Omar Marriott Hotel located in Makkah**. Tripadvisor is a widely used online travel platform where guests share their experiences and opinions about hotels, services, and facilities.

The dataset contains textual reviews written by guests describing their stay at the hotel. These reviews include feedback related to different aspects of the hotel such as **service quality, cleanliness, location, facilities, and overall customer satisfaction**. Each review represents a piece of unstructured text data that can be analyzed using text analytics techniques.

The reviews were collected from the Tripadvisor webpage of the hotel using **web data extraction with Selenium in Python**. The extracted fields include **reviewer name, review date, and review content**. After removing duplicate records, missing values, and very short reviews, the final dataset contains **497 English-language reviews**. The collected data is stored in **CSV format**, where each row represents one review entry.

However, there are some limitations in the dataset. Some reviews may be very short and contain limited information, while others may include informal language, spelling variations, or mixed writing styles. In addition, only English-language reviews are included in this project, which means that feedback written in other languages is excluded from the analysis.

5. Data Collection Method

The data for this project was collected from the Tripadvisor website, which contains publicly available reviews written by hotel guests. Tripadvisor was selected as the main data source because it provides real

customer opinions and experiences regarding hotel services, facilities, and overall satisfaction.

In this project, the reviews were collected using **Python with Selenium**, which allowed the extraction of guest review information directly from the hotel's Tripadvisor page. The extracted information includes the **reviewer name, review date, and review content**. The collected data was then saved in CSV format for preprocessing and further analysis.

6. Sampling

For this project, the study focuses on **497 English-language reviews** collected from the Tripadvisor page of Jabal Omar Marriott Hotel, Makkah. This sample is considered suitable for the scope of the project because it provides a sufficient amount of customer feedback to identify overall sentiment trends and common themes related to hotel services and guest experiences.

The selected reviews represent opinions from different guests and contain feedback on areas such as **service, cleanliness, location, and facilities**.

7. Text Preprocessing

Before performing sentiment analysis, the collected textual data must be cleaned and prepared. Text preprocessing is an important step in text analytics because raw text data often contains noise such as punctuation, URLs, numbers, inconsistent formatting, and common words that do not contribute meaningful information to the analysis.

The following preprocessing steps are applied in this project:

7.1 Lowercasing

All text data is converted into lowercase format to ensure consistency. This helps treat words such as *Hotel*, *hotel*, and *HOTEL* as the same word during analysis.

7.2 URL Removal

Any URLs or web links appearing in the reviews are removed because they do not contribute meaningful information to the sentiment analysis.

7.3 Removing Punctuation

Punctuation marks such as commas, periods, apostrophes, and exclamation marks are removed from the text because they do not contribute meaningful information to the analysis.

7.4 Removing Numbers

Numbers are removed from the review text because they usually do not provide significant meaning for text-based sentiment analysis.

7.5 Removing Extra Spaces

Extra spaces created during the cleaning process are removed to ensure the text is neat and standardized.

7.6 Stop Word Removal

Common words such as *the, is, a, and, to,* and *of* are removed because they occur very frequently in text but do not provide significant meaning for analysis.

7.7 Lemmatization

Lemmatization is applied to reduce words to their base or dictionary form. This helps group similar words together and improves the quality of text analysis.

7.8 Tokenization

Tokenization is the process of splitting sentences into individual words or tokens. This allows the text to be analyzed word by word.

Example:

Sentence:
“The hotel location is excellent.”

Tokens:
[“hotel”, “location”, “excellent”]

These preprocessing steps help transform the raw textual data into a cleaner and more structured format, which allows further analysis such as

sentiment analysis and topic identification to be performed more effectively.

8 Ethical Considerations

In this project, ethical considerations are taken into account when collecting and analyzing online review data. The reviews used in this study are publicly available on the Tripadvisor platform and are used only for academic purposes.

No personal or sensitive information about the reviewers is collected or analyzed. The analysis focuses only on the textual content of the reviews rather than the identities of the users.

Additionally, the dataset may contain subjective opinions from different individuals, which may introduce bias in the analysis. Therefore, the results of this study are interpreted carefully to ensure that conclusions are based on general patterns found in the data rather than individual opinions.

9. Dataset Source

Tripadvisor. (2024). Jabal Omar Marriott Hotel Makkah – Guest Reviews.

Available at: **https://www.tripadvisor.com.my/Hotel_Review-g293993-d8142637-Reviews-Jabal_Omar_Marriott_Hotel_Makkah-Mecca_Makkah_Province.html**